Fourier Tower Transforms and Quantum Circuits for Solvable Symmetries

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Abstract

We introduce a hierarchical framework for quantum Fourier analysis over finite solvable groups, called *Fourier Tower transforms*. Instead of directly constructing the full nonabelian quantum Fourier transform, we exploit a normal series decomposition of the group and build a layered circuit architecture adapted to the internal solvable structure of the symmetry group.

Each quotient layer is equipped with an abelian Fourier transform, while nonabelian structure is encoded through conjugation-induced controlled unitary operations coupling adjacent layers. The resulting transform defines a structured intermediate representation between classical abelian Fourier analysis and the full nonabelian quantum Fourier transform.

The construction naturally produces a hierarchy of interacting Fourier sectors, where the interaction between layers reflects the extension structure of the group. In this way, nonabelianity is interpreted not as a failure of Fourier analysis, but as a coupling phenomenon between compatible abelian Fourier resolutions.

This article extends the previous Fourier Tower circuit construction by adding a more detailed representation-theoretic, circuit-theoretic, extension-theoretic, and quantum-channel interpretation.

We develop the theory in detail, provide explicit constructions for the groups S_3 and S_4, discuss cocycle and extension structures, analyze quantum circuit complexity, and give a Qiskit-compatible implementation.

Finally, we interpret the Fourier Tower in three complementary viewpoints:

- as a layered quantum circuit,
- as a filtered representation-theoretic decomposition,
- and as a symmetry-resolved quantum channel.

1 Introduction

Quantum Fourier transforms over finite groups are fundamental tools in quantum computation, harmonic analysis, and symmetry-based algorithms.

They appear prominently in:

- hidden subgroup problems,
- phase estimation,
- quantum simulation,
- representation-theoretic quantum algorithms,
- and symmetry-adapted quantum information processing.

For finite abelian groups, the quantum Fourier transform admits an efficient implementation using one-dimensional characters:

$$\mathbb{C}[G] \cong \bigoplus_{\chi \in \widehat{G}} \mathbb{C}_\chi.$$

The abelian structure allows complete diagonalization into scalar Fourier modes.

For nonabelian groups, however, the situation becomes substantially more complicated. The group algebra decomposes into matrix blocks:

$$\mathbb{C}[G] \cong \bigoplus_{\rho \in \widehat{G}} V_\rho \otimes V_\rho^*,$$

where:

- $\widehat{G}$ is the set of irreducible representations,
- V_ρ is the representation space of ρ.

Direct implementation of this decomposition is often difficult:

- irreducible representations may be high-dimensional,
- representation multiplicities may become complicated,
- Clebsch–Gordan structures may appear,
- efficient circuit decompositions are generally nontrivial.

The goal of the present work is not to replace the full nonabelian quantum Fourier transform. Rather, the goal is to propose a complementary framework adapted to solvable groups.

1.1 Relation with the Previous Fourier Tower Construction

This article should be viewed as a continuation and refinement of the earlier Fourier Tower circuit construction.

The previous version introduced the basic idea:

$$\text{normal series} \quad \longrightarrow \quad \text{quotient Fourier layers} \quad \longrightarrow \quad \text{controlled conjugation gates.}$$

The present version adds:

- a clearer role for transversal coordinates,
- a more precise tensor interpretation,
- a semidirect-product viewpoint,
- a cocycle and extension interpretation,
- a quantum-channel interpretation,
- and a more detailed relation with the full nonabelian Fourier transform.

Thus the original Fourier Tower circuit may be regarded as the operational realization of a broader solvable-group Fourier framework.

1.2 Main Idea

Instead of directly diagonalizing the full nonabelian group algebra, we propose a different viewpoint.

If the group is solvable, then it admits a normal series:

$$\{e\} = G_0 \triangleleft G_1 \triangleleft \cdots \triangleleft G_n = G$$

with abelian quotients:

$$Q_i = G_{i+1}/G_i.$$

The central idea of this paper is:

$$\boxed{\text{nonabelian structure} = \text{interaction between Fourier layers}}$$

Rather than performing one global Fourier transform, we:

1. perform local abelian Fourier transforms on each quotient layer,
2. then propagate symmetry information between layers using conjugation-induced controlled operations.

This produces a hierarchical transform called the:

Fourier Tower Transform.

1.3 Philosophical Interpretation

The Fourier Tower should not be viewed as a replacement for the full nonabelian QFT, but rather as:

- an intermediate hierarchical transform,
- a symmetry-adapted decomposition,
- and a structured propagation of Fourier information across subgroup levels.

The construction reveals that solvable nonabelian groups possess a layered Fourier geometry.

1.4 Elementary Fourier Principle

The Fourier Tower construction may be viewed as a hierarchical generalization of the elementary identity:

$$HXH = Z,$$

where:

- H is the Hadamard transform,
- X is the translation operator on $\mathbb{Z}_2$,
- and Z is the corresponding phase operator.

In this elementary situation, the Fourier transform converts translation symmetry into phase symmetry.

The Fourier Tower extends this principle from a single abelian symmetry layer to a hierarchy of interacting quotient symmetries.

2 Normal Series and Tower Coordinates

Definition 1 (Normal Series)**.** *A normal series of a finite group G is a sequence:*

$$\{e\} = G_0 \triangleleft G_1 \triangleleft \cdots \triangleleft G_n = G.$$

Each quotient:

$$Q_i = G_{i+1}/G_i$$

defines a structural layer.

If G is solvable, all quotients Q_i are abelian.

2.1 Transversal Coordinates

Definition 2 (Transversal Set)**.** *Let:*

$$H \triangleleft G.$$

A transversal set for the quotient G/H is a subset:

$$T \subset G$$

containing exactly one representative from each coset of H.

Once such a transversal is fixed, every element $g \in G$ admits a unique decomposition:

$$g = th, \qquad t \in T, \quad h \in H.$$

For a subgroup tower:

$$\{e\} = G_0 \triangleleft \cdots \triangleleft G_n = G,$$

choose transversal sets:

$$T_i \subset G_{i+1}$$

for each quotient:

$$G_{i+1}/G_i.$$

Then every group element admits a decomposition:

$$g = t_{n-1}t_{n-2}\cdots t_0, \qquad t_i \in T_i.$$

These coordinates define a hierarchical factorization of group elements.

2.2 Tensor Structure

The tower decomposition induces a vector-space identification:

$$\mathbb{C}[G] \simeq \bigotimes_i \mathbb{C}[Q_i].$$

Equivalently, basis states may be written as:

$$|g\rangle = |t_{n-1}\rangle \otimes \cdots \otimes |t_0\rangle.$$

Importantly, this is not generally an algebra isomorphism:

- tensor factors interact through conjugation,
- extension data couples layers,
- multiplication is twisted.

Thus the tensor decomposition should be understood as:

$$\text{local Fourier structure} + \text{global coupling}.$$

3 Fourier Tower Construction

3.1 Local Fourier Layers

Definition 3 (Fourier Layer)**.** *For each abelian quotient Q_i, define:*

$$F_{Q_i} : \mathbb{C}[Q_i] \to \bigoplus_{\chi \in \widehat{Q_i}} \mathbb{C}_\chi.$$

Since each Q_i is abelian, these transforms are ordinary abelian Fourier transforms.

For cyclic quotients:

$$Q_i \cong \mathbb{Z}_n,$$

the transform is the usual discrete Fourier transform.

3.2 Conjugation Structure

The higher layers act on lower layers through conjugation.

Definition 4 (Conjugation Action)**.** *Let y be an element represented in a higher quotient layer. Conjugation induces an action on a lower quotient layer:*

$$\sigma_y(x) = yxy^{-1}.$$

In quotient notation, this should be interpreted through representatives of cosets. The induced action is well-defined whenever the corresponding normality and compatibility conditions are satisfied.

This induces permutation operators:

$$P_y|x\rangle = |\sigma_y(x)\rangle.$$

3.3 Controlled Symmetry Gates

Definition 5 (Controlled Conjugation Gate)**.** *The controlled conjugation gate associated with adjacent layers is:*

$$C_i = \sum_{y \in Q_{i+1}} |y\rangle\langle y| \otimes P_y.$$

Thus:

$$C_i : |y\rangle|x\rangle \mapsto |y\rangle|\sigma_y(x)\rangle.$$

These operators propagate symmetry information between adjacent Fourier layers.

4 Fourier Tower Transform

Definition 6 (Fourier Tower Transform)**.** *The Fourier Tower transform associated with the normal series $\mathcal{T}$ is:*

$$F_{\mathcal{T}} = \left(\prod_i C_i\right)\left(\bigotimes_i F_{Q_i}\right).$$

4.1 Interpretation

The transform proceeds in two stages:

1. Independent abelian Fourier analysis inside each quotient layer,
2. Symmetry propagation through controlled conjugation.

Thus:

$$\text{nonabelianity} = \text{interaction between local Fourier sectors.}$$

5 Fundamental Theorem

Theorem 1 (Unitarity of the Fourier Tower Transform)**.** *Let G be a finite solvable group with normal series:*

$$\{e\} = G_0 \triangleleft \cdots \triangleleft G_n = G.$$

Assume the quotient-layer actions are implemented by permutation operators induced by conjugation. Then the Fourier Tower transform:

$$F_{\mathcal{T}} = \left(\prod_i C_i\right)\left(\bigotimes_i F_{Q_i}\right)$$

defines a unitary operator on $\mathbb{C}[G]$.

Proof. Each local Fourier transform F_{Q_i} is unitary because Q_i is finite abelian.

Each controlled conjugation gate C_i is a controlled permutation operator. Since each P_y is a permutation unitary, C_i is unitary.

A composition of unitary operators is unitary.

Hence:

$$F_{\mathcal{T}}$$

is unitary. □

6 Example: The Group S_3

The symmetric group S_3 admits:

$$S_3 \cong \mathbb{Z}_3 \rtimes \mathbb{Z}_2.$$

6.1 Layer Structure

We take:

$$Q_0 = \mathbb{Z}_3, \qquad Q_1 = \mathbb{Z}_2.$$

Equivalently:

$$\{e\} \triangleleft A_3 \triangleleft S_3,$$

with:

$$A_3 \cong \mathbb{Z}_3, \qquad S_3/A_3 \cong \mathbb{Z}_2.$$

6.2 Conjugation Action

Let:

$$r = (123), \qquad s = (12).$$

Then:

$$srs^{-1} = r^{-1}.$$

Thus the $\mathbb{Z}_2$ layer exchanges Fourier modes:

$$\chi_1 \leftrightarrow \chi_2.$$

6.3 Tower Transform

The Fourier Tower becomes:

$$F_{\mathcal{T}} = C(F_{\mathbb{Z}_2} \otimes F_{\mathbb{Z}_3}).$$

The nonabelian structure appears entirely through controlled conjugation.

7 Example: The Group S_4

A solvable decomposition of S_4 is:

$$\{e\} \triangleleft V_4 \triangleleft A_4 \triangleleft S_4.$$

Equivalently, at the level of quotient structure:

$$S_4 \cong (V_4 \rtimes \mathbb{Z}_3) \rtimes \mathbb{Z}_2.$$

7.1 Layer Structure

The layers are:

$$V_4 \cong \mathbb{Z}_2 \times \mathbb{Z}_2, \qquad A_4/V_4 \cong \mathbb{Z}_3, \qquad S_4/A_4 \cong \mathbb{Z}_2.$$

7.2 Interpretation

The $\mathbb{Z}_3$ action cyclically permutes Fourier sectors of V_4.

The resulting transform has:

- local Fourier analysis,
- cyclic symmetry propagation,
- layered entanglement.

8 Quantum Circuit Realization

Definition 7 (Fourier Tower Circuit)**.**

$$U_G = \left(\prod_i C_i\right)\left(\bigotimes_i F_{Q_i}\right).$$

8.1 Layered Architecture

The circuit admits a natural hierarchy:

- bottom layer: abelian Fourier transforms,
- middle layer: controlled conjugation entanglement,
- top layer: global symmetry propagation.

8.2 Interpretation

The group structure determines:

- register decomposition,
- qubit or qudit connectivity,
- control structure,
- and entangling geometry.

9 Qiskit-Compatible Implementation for S_3

The following code gives a simple Qiskit-compatible illustration of a Fourier Tower circuit for the S_3 structure.

It should be understood as a pedagogical circuit skeleton rather than a full optimized implementation of the nonabelian Fourier transform on S_3.

```
from qiskit import QuantumCircuit
import numpy as np

# Three-qubit illustrative register:
# qubits 0 and 1 encode a Z_3-like layer approximately,
# qubit 2 encodes the Z_2 layer.
qc = QuantumCircuit(3)

# Fourier transform on the Z_2 quotient layer
qc.h(2)

# Simple illustrative Fourier-like operations on the lower layer
qc.h(0)
qc.cp(np.pi/2, 0, 1)
qc.h(1)

# Controlled conjugation action from Z_2 to the lower layer
qc.cx(2, 0)
qc.cx(2, 1)

qc.measure_all()
```

Remark 1. *A fully faithful implementation of the S_3 Fourier Tower should encode the $\mathbb{Z}_3$ layer using a qutrit or an appropriate two-qubit embedding with unused states. The circuit above is meant to illustrate the architecture: local Fourier gates followed by controlled symmetry gates.*

10 Three Complementary Viewpoints on Fourier Towers

The Fourier Tower admits three complementary viewpoints.

Each viewpoint describes the same mathematical object:

$$F_{\mathcal{T}} : \mathbb{C}[G] \to \mathbb{C}[G].$$

10.1 Structural Equivalence Principle

$$\text{Circuit} \longleftrightarrow \text{Representation} \longleftrightarrow \text{Quantum Channel}.$$

10.2 Option A: Circuit-Theoretic Viewpoint

In this viewpoint:

$$U_G = \left(\prod_i C_i\right)\left(\bigotimes_i F_{Q_i}\right)$$

is interpreted as a physical quantum circuit.

10.2.1 Operational Interpretation

1. local Fourier analysis,
2. symmetry propagation,
3. entangling nonabelian structure.

10.2.2 Meaning

Here:

- group action becomes gate connectivity,
- conjugation becomes entanglement,
- Fourier layers become computational subsystems.

10.3 Option B: Representation-Theoretic Viewpoint

The Fourier Tower is interpreted as a filtered representation decomposition.

As a vector space:

$$\mathbb{C}[G] \simeq \bigotimes_i \mathbb{C}[Q_i].$$

10.3.1 Twisted Representation Structure

The full representation is not generally:

$$\rho_G = \bigotimes_i \rho_i.$$

Instead, it is more accurately viewed as a tensor structure modified by conjugation twisting:

$$\rho_G \sim \left(\bigotimes_i \rho_i \right) \quad \text{with conjugation-induced coupling.}$$

10.3.2 Interpretation

The Fourier Tower does not fully diagonalize G.

Instead it produces:

- local diagonalization,
- global coupling,
- filtered symmetry propagation.

10.4 Option C: Quantum Channel Viewpoint

Let:

$$\widetilde{\rho} = F_{\mathcal{T}} \rho F_{\mathcal{T}}^{\dagger}.$$

If hidden layers are ignored:

$$\Phi(\rho) = \mathrm{Tr}_{\text{hidden}}(\widetilde{\rho}).$$

10.4.1 Kraus Operators

The channel admits:

$$\Phi(\rho) = \sum_k K_k \rho K_k^{\dagger},$$

with:

$$K_k = (\langle k| \otimes I) F_{\mathcal{T}}.$$

10.4.2 Physical Meaning

Each Kraus operator corresponds to:

- selecting a Fourier layer,
- discarding hidden symmetry sectors,
- inducing structured decoherence.

11 Representation-Theoretic Interpretation

11.1 Symmetry Group of an Operator

Let:

$$U \in \mathcal{U}(\mathcal{H})$$

be a unitary operator.

Suppose a finite group:

$$G_U$$

acts on $\mathcal{H}$ through:

$$\rho : G_U \to \mathcal{U}(\mathcal{H}),$$

and assume:

$$U\rho(g) = \rho(g)U.$$

Then G_U defines a symmetry group of the operator U.

11.2 Fourier Representation

Define:

$$\widetilde{U} = F_{\mathcal{T}} U F_{\mathcal{T}}^{-1}.$$

The Fourier Tower does not necessarily fully diagonalize U.

Instead it organizes the operator into:

- local Fourier sectors,
- coupled symmetry layers,
- and filtered representation structures.

Remark 2. *The Fourier Tower may therefore be interpreted as a hierarchical generalization of elementary Fourier identities such as:*

$$HXH = Z.$$

For elementary gates such as CNOT or Toffoli, the symmetry group is abelian.

The Fourier Tower extends this principle to solvable nonabelian symmetries.

12 Relation with the Full Nonabelian Fourier Transform

The full nonabelian Fourier transform decomposes:

$$\mathbb{C}[G] \cong \bigoplus_{\rho \in \widehat{G}} V_\rho \otimes V_\rho^*.$$

Explicitly:

$$F_G|g\rangle = \bigoplus_{\rho \in \widehat{G}} \sqrt{\frac{d_\rho}{|G|}} \sum_{i,j} \rho(g)_{ij} |\rho, i, j\rangle.$$

12.1 Difference with Fourier Towers

The Fourier Tower does not directly isolate all irreducible representations.

Instead it constructs:

- local abelian Fourier decompositions,
- controlled conjugation couplings,
- and hierarchical symmetry propagation.

Thus the Fourier Tower should be interpreted as:

- a structured intermediate transform,
- a solvable-group-adapted Fourier framework,
- and a layered representation decomposition.

13 Group Extension and Cocycle Structure

Each layer extension fits into:

$$1 \to G_i \to G_{i+1} \to Q_i \to 1.$$

13.1 Cocycle Structure

After choosing a section of the quotient map, extension data may be described by maps:

$$\omega_i : Q_i \times Q_i \to G_i.$$

Group multiplication can be written schematically as:

$$(x, a)(y, b) = (x\alpha_a(y)\omega_i(a, b), ab).$$

13.2 Interpretation in Fourier Towers

The Fourier Tower implements:

- α_a through controlled conjugation gates,
- ω_i through twisting interactions between layers.

Thus the Fourier Tower may be interpreted as a unitary construction compatible with solvable extension data.

14 Main Extension Theorem

Theorem 2 (Fourier Tower and Solvable Extensions)**.** *Let:*

$$\{e\} = G_0 \triangleleft \cdots \triangleleft G_n = G$$

be a solvable normal series.

Then:

$$F_{\mathcal{T}} = \left(\prod_i C_i\right)\left(\bigotimes_i F_{Q_i}\right)$$

provides a unitary Fourier-layer representation compatible with the twisted tensor-product structure determined by the extension data of the series.

Proof Sketch. Each quotient contributes a local abelian Fourier transform.

The extension data determine the twisting between layers.

Controlled conjugation operators implement the group action between adjacent quotient layers.

Thus the full transform encodes the solvable extension structure at the level of a unitary layered Fourier representation. □

15 Complexity of Fourier Tower Circuits

Proposition 1. *Let G be solvable with quotients Q_i.*

Then the Fourier Tower circuit has complexity:

$$O\left(\sum_i \mathrm{cost}(F_{Q_i}) + \sum_i \mathrm{cost}(C_i)\right).$$

Remark 3. *If:*

$$Q_i \cong \mathbb{Z}_{n_i},$$

then each cyclic Fourier transform may be implemented with cost:

$$O(\log^2 n_i).$$

The remaining complexity is governed by the controlled conjugation operators.

Remark 4. *The gain comes from avoiding direct decomposition into irreducible representations of the full group.*

Instead, the computation is organized through abelian Fourier layers and controlled symmetry interactions.

16 Physical Interpretation

The Fourier Tower induces:

- structured decoherence,
- symmetry-resolved entanglement,
- controlled permutation of Fourier sectors,
- hierarchical propagation of quantum information.

From this viewpoint, subgroup layers behave as symmetry registers. Fourier transforms resolve local symmetry information, while conjugation gates propagate this information through the hierarchy.

17 Conclusion

We introduced Fourier Tower transforms as a hierarchical framework for quantum Fourier analysis on solvable groups.

The main principle is:

$$\boxed{\text{nonabelian structure} = \text{interaction between Fourier layers}}$$

The construction provides simultaneously:

- a quantum circuit architecture,
- a representation-theoretic decomposition,
- a cocycle-extension framework,
- and a quantum-channel interpretation.

The Fourier Tower suggests a new approach to symmetry-adapted quantum computation based on layered propagation of Fourier information.

Future work includes:

- rigorous links with Clifford and Mackey theory,
- explicit implementations for larger solvable groups,
- tensor-network interpretations,
- optimized qudit and qubit circuit realizations,
- and applications to symmetry-adapted quantum algorithms.

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